An STS Case Study on Digital Privacy and Social Media

Science, Technology and Society

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I. Introduction

Background of the Issue

In the era of 21st century, social media is considered as one of the important technological tools used for human communication and interaction. Social media platforms such as Facebook, Instagram, TikTok, and X known as Twitter way back then, have changed the way we connect with each other, sharing thoughts and forming relationships. These products promote open communication and self-disclosure, permitting people to share private information like the photos, opinions, and location, with a worldwide audience.

But this added interconnectivity raises fears over the securitization of data. Every post, like and comment is written into the company's digital storehouse of information that can be kept, handed off to someone else or even sold, again without the user's explicit agreement. Broad use of algorithms, artificial intelligence and data analytics means people's personal online behavior is being tracked continuously, usually for marketing, but sometimes to influence elections.

Digital privacy has emerged as a major issue worldwide particularly as users are not actually aware how their data is taken and used. Incidents like identity theft, online scams and even the Cambridge Analytica scandal recall how our personal data can be turned into a weapon to influence behaviors and manipulate public opinion. This is particularly true of adolescents and young adults, those most prolific users of social media and who are frequently unaware of the potential dangers associated with "sharing too much." As technology continues to evolve, the protection of personal data remains one of the most pressing ethical and social challenges of our time.

Relevance to Science, Technology, and Society (STS):

The problem of privacy and social media is an example for the tight interrelation between scientific developments, technological progress and social behavior, the centre of attention in Science, Technology and Society (STS) research. From the increasingly wide scope of computational abilities and power, tools have been developed that change what can be accomplished by people in thinking, feeling or acting. These very same tools, however, have also raised ethical and cultural concerns of trust, consent, autonomy and surveillance.

In this sense, under an STS framework social media is not simply a technology but also a social artefact molded by human values and decisions. And it shows how the cultural expectations and political environment that science emerging from society, at the same time produces new social norms. For instance, issues of data privacy are one aspect of how unbridled technological expansion can infringe on personal rights, ethical norms and digital citizenship. Through examining this issue we can gain a clearer understanding of the social responsibilities for technological progress and ethical responsibility in a digital era.

Statement of the Problem:

How does the use of social media platforms affect users' data privacy, and what are the social, ethical, and technological implications of sharing personal information online?

Purpose and Objectives:

The objective of this case is to discuss digital privacy when using social media and illustrate the ethical dimensions that are brought about by technological innovation in data collection and sharing, from individual and societal perspective.

Specifically, the objectives are:

1. To investigate how social media sites gather, store and utilize data from users.
2. To establish the social, moral, and psychological effects of privacy losses on digital platforms.
3. To examine the contribution of legislation, policy and education in the promotion of responsible use online.
4. To propose some possible directions or solutions to enhance the awareness and protection of data privacy.

II. Case Description

Case Overview:

One of the largest known real-world examples of violating social media privacy occurred in 2018 with the Cambridge Analytica scandal. This British political consulting firm acquired the personal information of about 87 million Facebook users through a personality quiz app without their consent. The information was used to produce psychological profiles of people and then deliver appropriate political ads during campaigns, such as the 2016 U.S. Presidential campaign.

The scandal underscored how easy it was to hijack personal information shared online, capitalizing on it for politics and business. It raised serious ethical questions about how responsible tech companies are for protecting user data and making their algorithms transparent. After public outcry, Facebook came under world scrutiny and was fined billions of dollars for encroaching on users' privacy.

Privacy issues in the Philippines have also led to a significant raise as online scams, phishing attacks and fake accounts keep popping. The Data Privacy Act of 2012 (Republic Act

No. 10173) was specifically crafted to safeguard data, but its enforcement is challenging, with tech advancements constantly outpacing the law.

Scientific or Technological Background:

Social networks make extensive use of data analytics, machine learning and AI to improve user experience. These technologies gather information on users' actions—like locations, clicks or preferences—to create customized feeds and ads. While this flow is used to generate appealing content, it contributes additionally to data surveillance.

Algorithms read user interest and dictate what people see online, in platforms that produce an "echo chamber" for those views. The very tools that make social media so addictive also make it more accessible for companies to track and profile people — privacy and ethical considerations about issues of consent, control and manipulation.

Key Stakeholders:

- **Social Media Users:** The primary individuals affected by privacy breaches, often unaware of the risks.
- **Social Media Companies:** Including the Meta, Tiktok, X and etc. are responsible for managing and securing user data.
- **Government Agencies:** Tasked with enforcing data protection laws and regulations.
- **Cybersecurity Experts and Researchers:** Develop tools and frameworks to prevent data misuse.
- **Society as a Whole:** Faces the cultural and ethical consequences of digital surveillance and data manipulation.

III. Review of Related Literature

The phenomenon of data privacy concerns relative to social media has been echoed across various scholarly realms from politics to sociology, to research endeavors relatively to social media's impact on user agency, security, and trust in the digital world. A variety of studies exist addressing personal data collection and processing, exploitation and user response efforts.

In The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power, Zuboff (2019) investigates the concept of surveillance capitalism. She finds that major technology companies like Facebook and Google rely on their user's private information to profit and that personal user experience becomes the new raw material for capitalistic gain. They accumulate such data and then through an extraction process they keep hidden, manage predictive analytics and user influence. Ultimately, Zuboff (2019) cites similar patterns of exploitation against personal privacy that undermine democratic institutions and capitalist freedom; therefore, this commodification of the personal, otherwise private, should be approached with an understanding that data privacy is not only a technical concern but a sociological and ethical one as well.

Similarly, the Pew Research Center (2019) conducted a comprehensive study titled Americans and Privacy: Concerned, Confused, and Feeling Lack of Control Over Their Personal Information, which reveals that most social media users are deeply worried about their privacy but feel powerless to control their data. The report highlights a paradox between awareness and action, people recognize privacy risks but continue to engage in behaviors that expose their information online. This reflects a growing sense of “privacy resignation,” where users accept data tracking as an unavoidable aspect of digital life. The study emphasizes the need for stronger privacy education and clearer data protection regulations to restore users’ sense of control.

Tufekci (2015), in her article Algorithmic Harms Beyond Facebook and Google: Emergent Challenges of Computational Agency, examines how social media algorithms manipulate users' experiences by filtering and prioritizing information. She explains that algorithmic systems are designed to maximize engagement but can unintentionally (or deliberately) amplify biases, misinformation, and emotional manipulation. Tufekci raises ethical concerns about the lack of transparency in how algorithms operate, arguing that users often have little understanding of how their online environments are shaped by unseen computational forces. Her work underscores the importance of algorithmic accountability and the ethical responsibilities of technology developers.

Taken together, these studies present complementary perspectives on the complex issue of digital privacy. Zuboff (2019) focuses on the economic and structural aspects of data exploitation, Pew Research Center (2019) provides empirical insights into public attitudes and behaviors, and Tufekci (2015) highlights the technological and ethical dimensions of algorithmic influence. However, a gap remains in the literature regarding how teenagers and young adults, who are among the most active users of social media, understand and manage their privacy online. This case study seeks to address this gap by exploring the awareness, behavior, and perceptions of younger users in relation to digital privacy, as well as identifying strategies to promote safer and more informed online practices.

IV. Analysis and Discussion

STS Framework or Lens:

This study applies the Ethics of Technology and Technological Determinism frameworks. The Ethics of Technology framework examines how innovations create moral challenges for society, particularly regarding privacy, freedom, and accountability.

Technological Determinism, meanwhile, explores how technological tools influence human behavior and societal change, often faster than ethical norms can adapt.

Social and Ethical Dimensions:

The social implications of data privacy violations include loss of trust, exposure to scams, and online harassment. Ethically, it raises questions about consent, transparency, and surveillance. Culturally, social media influences how individuals present themselves online, often prioritizing popularity over privacy. Teenagers and young adults are most vulnerable, as they tend to share personal details without understanding long-term consequences.

Interaction Between Science/Technology and Society:

Technology and society influence each other continuously. Social media technologies shape user behavior through algorithms and recommendation systems, while public demand for transparency and ethical conduct pushes companies and governments to adapt privacy policies. The rise of the Data Privacy Act of 2012 in the Philippines and the General Data Protection Regulation (GDPR) in Europe are examples of society responding to technological change through legal frameworks.

Challenges and Conflicts:

The case presents several challenges and conflicts, including the difficulty of balancing innovation and privacy in social media design, weak enforcement of data protection laws, users' lack of awareness about online safety, companies prioritizing profit over ethical

responsibility, and the challenges of achieving global privacy standards in an interconnected world.

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